Business analysis report

Planted sample

Version	1
Data period	2023-01-06 – 2024-12-31
Rows	21,971
Data-quality score	100 / 100
Currency	not stated
Analysis run	2026-11-05
Generated	2026-11-05

> Every figure in this report comes from the stored results of the analysis run named above: SQL and statistical models that ran on your data. Nothing was estimated or written by an AI model. What-if scenarios are not included.

Executive summary

The findings with the largest estimated business impact, highest first.

1. Year to date is 196.7% ahead of last year
Net revenue from 2024-01-01 to 2024-12-31 is 9,112,029.74, versus 3,071,272.26 for 2023-01-01 to 2023-12-31 (196.7%).
Confidence: High · Sample: 21,971 · Impact score: 100

2. Lasting downward shift in orders from 2024-03-11
Weekly orders stepped down by about 46.8% from the week of 2024-03-11 and stayed at the new level for 23 weeks (compared with the same weeks a year earlier, so the normal season is taken out). That is roughly -2,574,048.61 of revenue since. These are findings from the data, not proven causes.
Confidence: High · Sample: 103 · Impact score: 84

3. Net revenue for Furniture is estimated at about 1,339,349.34 next month (80% range: 588,591.74 to 2,090,106.93)
It is an estimate, not a promise: Net revenue for Furniture is estimated at about 1,339,349.34 for Jan 2025 (80% range: 588,591.74 to 2,090,106.93; 95% range: 191,164.68 to 2,487,534.00). Over the next 6 months the estimated total is about 8,603,653.42. Method: exponential smoothing (ETS), chosen by testing on past data (3 rolling windows of 5 months). In those tests its estimates were off by about 14.9% on average (sMAPE) and by about 89,093.02 per period (MAE); its error was 0.32 times that of repeating the previous cycle (MASE, below 1 is better). The seasonal naive baseline was off by about 96.4% (MASE 1.07). Treat it as a guide rather than a firm number.
Confidence: Medium · Sample: 7,002 · Impact score: 60

4. Net revenue is estimated at about 2,295,213.14 next month (80% range: 1,105,837.97 to 3,484,588.30)
It is an estimate, not a promise: net revenue is estimated at about 2,295,213.14 for Jan 2025 (80% range: 1,105,837.97 to 3,484,588.30; 95% range: 476,220.79 to 4,114,205.48). Over the next 6 months the estimated total is about 14,674,100.06. Method: exponential smoothing (ETS), chosen by testing on past data (3 rolling windows of 5 months). In those tests its estimates were off by about 14.6% on average (sMAPE) and by about 153,425.60 per period (MAE); its error was 0.30 times that of repeating the previous cycle (MASE, below 1 is better). The seasonal naive baseline was off by about 98.4% (MASE 1.07). Treat it as a guide rather than a firm number.
Confidence: Medium | Sample: 21,971 | Impact score: 60

5. Champions customers bring 50.8% of revenue
1,363 customers fall into 7 RFM segments. Champions bring the most revenue: 50.8% from 328 customers. 437 customers are At risk or Hibernating; they used to spend about 1,751,622.83 a year. Suggestion: Win-back offer: a personal reminder or discount before they lapse for good. Segments describe how customers behaved; they do not explain why.
Confidence: High | Sample: 1,363 | Impact score: 58

Key figures

The dashboard's default view: the last 90 days of data (2024-10-03 to 2024-12-31) compared with the 90 days before (2024-07-05 to 2024-10-02).

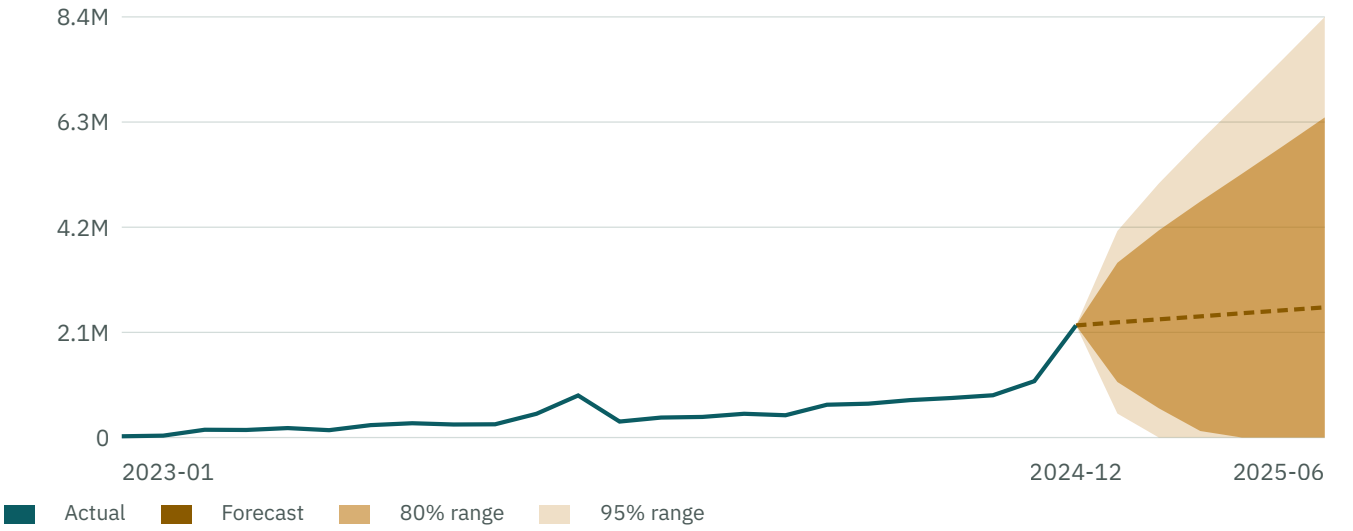
Confidence: High | Sample: 7,027

Measure	Value	Previous period	Change
Net revenue	4,155,778	2,189,144	+89.8%
Orders	7,025	4,111	+70.9%
Average order value	592	533	+11.1%
Units	15,114	8,671	+74.3%
Gross margin	38.1%	38.1%	0.0 pts

Trend and forecast

Net revenue per month with the forecast of the next periods. The darker band is the 80% range, the lighter band the 95% range. Model: AutoETS.

Confidence: Medium | Sample: 21,971



Period	Actual	Forecast	80% range	95% range
Jan 2024	319,366	–	–	–
Feb 2024	397,297	–	–	–
Mar 2024	410,730	–	–	–
Apr 2024	473,936	–	–	–
May 2024	445,315	–	–	–
Jun 2024	652,861	–	–	–
Jul 2024	675,644	–	–	–
Aug 2024	747,571	–	–	–
Sep 2024	786,578	–	–	–
Oct 2024	843,975	–	–	–
Nov 2024	1,123,625	–	–	–
Dec 2024	2,235,130	–	–	–
Jan 2025	–	2,295,213	1,105,838 – 3,484,588	476,221 – 4,114,205
Feb 2025	–	2,355,401	584,078 – 4,126,724	0 – 5,064,406
Mar 2025	–	2,415,589	130,414 – 4,700,764	0 – 5,910,463
Apr 2025	–	2,475,777	0 – 5,255,981	0 – 6,727,731
May 2025	–	2,535,965	0 – 5,811,919	0 – 7,546,104
Jun 2025	–	2,596,154	0 – 6,379,362	0 – 8,382,071

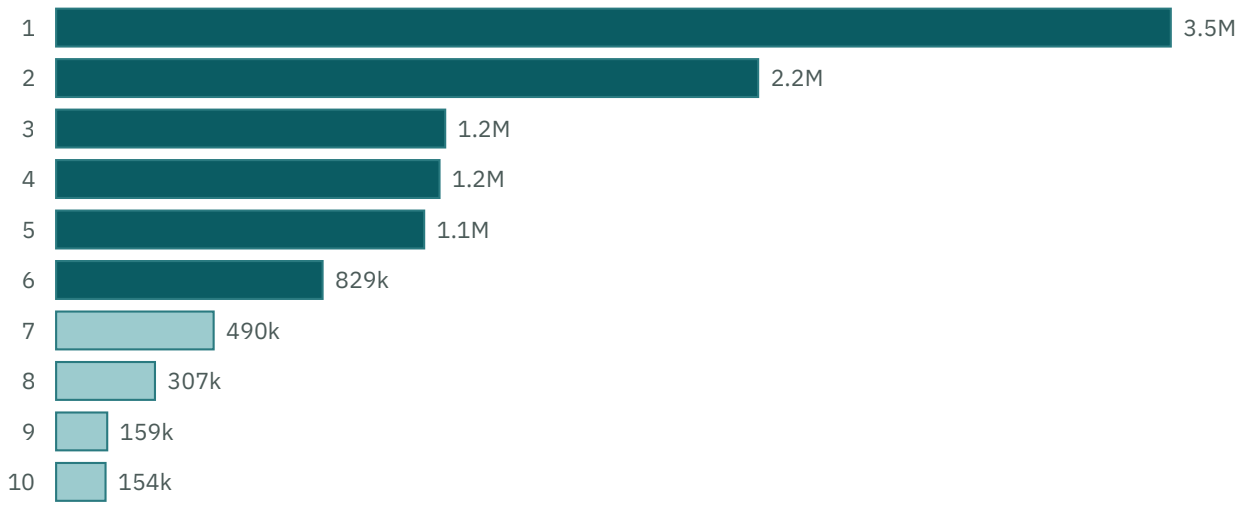
Top contributors

Positive sales only, over the whole file. A items make up the first 80% of sales, B items the next 15%, C items the rest.

Products

Confidence: High · Sample: 21,967

A: 6 · B: 8 · C: 16 of 30 items

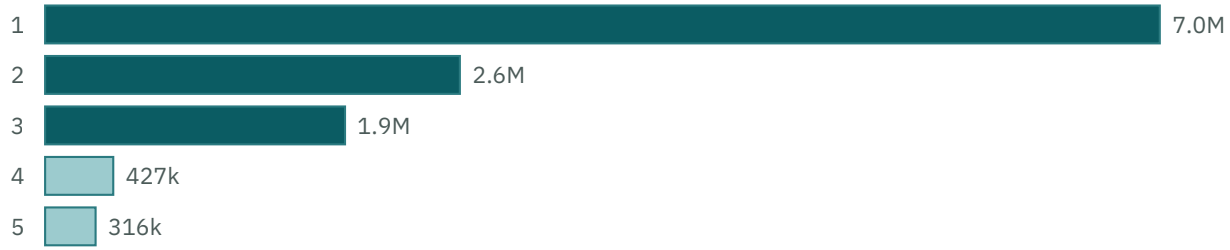


#	Item	Sales	Share	Cumulative	Class
1	Item-07	3,470,127	28.5%	28.5%	A
2	Item-12	2,185,541	17.9%	46.4%	A
3	Item-06	1,211,597	9.9%	56.4%	A
4	Item-19	1,193,537	9.8%	66.2%	A
5	Item-04	1,145,827	9.4%	75.6%	A
6	Item-02	829,316	6.8%	82.4%	A
7	Item-16	490,306	4.0%	86.4%	B
8	Item-27	307,403	2.5%	88.9%	B
9	Item-13	159,285	1.3%	90.2%	B
10	Item-17	153,745	1.3%	91.5%	B

Categories

Confidence: High · Sample: 21,967

A: 3 · B: 1 · C: 1 of 5 items

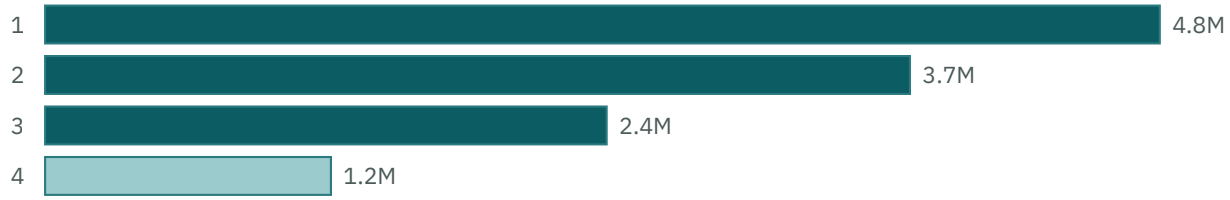


#	Item	Sales	Share	Cumulative	Class
1	Furniture	6,970,990	57.2%	57.2%	A
2	Grocery	2,593,395	21.3%	78.5%	A
3	Electronics	1,876,158	15.4%	93.9%	A
4	Office	426,768	3.5%	97.4%	B
5	Apparel	315,991	2.6%	100.0%	C

Regions

Confidence: High · Sample: 21,967

A: 3 · B: 1 · C: 0 of 4 items



#	Item	Sales	Share	Cumulative	Class
1	North	4,803,671	39.4%	39.4%	A
2	South	3,727,294	30.6%	70.0%	A

#	Item	Sales	Share	Cumulative	Class
3	East	2,420,198	19.9%	89.9%	A
4	West	1,232,139	10.1%	100.0%	B

Products without recent sales

2 of 30 products had no sales in the 90 days up to 2024-12-31.

Item-30, Item-29

Unusual days

Periods where net revenue was far from its usual level. 8 were found; the 8 largest are listed by date.

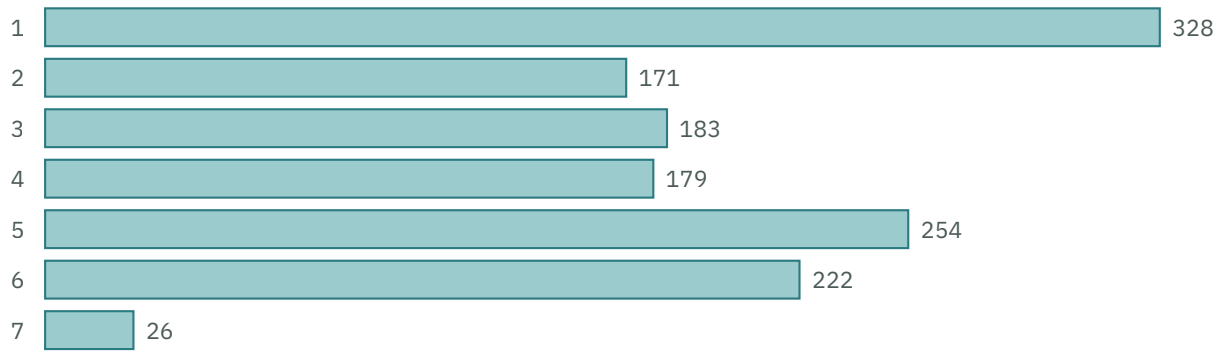
Confidence: High Sample: 726

Start	End	Kind	Actual	Usual level	Largest changes in
2023-01-09	2023-01-09	Spike	2,696	376	South, Item-07, Furniture
2023-01-27	2023-01-27	Drop	0	36.77	South, North, Item-04
2023-01-29	2023-01-29	Spike	3,429	32.26	East, North, Item-07
2023-02-17	2023-02-17	Drop	0	771	South, North, Item-07
2023-03-14	2023-03-14	Spike	45,812	2,164	North, South, Item-07
2023-05-09	2023-05-09	Drop	37.50	7,018	North, South, Item-07
2023-10-17	2023-10-17	Drop	9.98	6,457	North, South, Item-07
2024-02-20	2024-02-20	Spike	126,180	9,512	North, South, Item-07

Customer segments

1,363 customers grouped by how recently, how often and how much they buy. Actions are suggestions, not tested results.

Confidence: High Sample: 1,363



#	Segment	Customers	Share of sales	Suggested action
1	Champions	328	50.8%	Suggestion: Reward them: early access, loyalty perks, and ask for reviews and referrals.

#	Segment	Customers	Share of sales	Suggested action
2	Loyal	171	17.0%	Suggestion: Upsell higher-value products and invite them to a loyalty programme.
3	At risk	183	13.4%	Suggestion: Win-back offer: a personal reminder or discount before they lapse for good.
4	Potential loyalists	179	9.5%	Suggestion: Give a second-purchase incentive or membership to build the habit.
5	Hibernating	254	5.3%	Suggestion: Low-cost reactivation with seasonal offers; do not over-invest.
6	Lost	222	3.6%	Suggestion: Try one win-back campaign; otherwise stop spending marketing money on them.
7	New	26	0.3%	Suggestion: Onboard well: a welcome series and a reason to buy a second time soon.

Churn risk

1,170 customers were scored for the chance they stop buying within 32 days. Only totals are shown; no customer is named.

Confidence: High Sample: 1,170

Group	Customers	Yearly spend	Expected loss
High risk	355	1,473,439	1,279,529
Medium risk	111	–	–
Low risk	704	–	–
Top 10% by risk	117	543,452	540,734

Assumptions, caveats and data quality

Revenue is net of returns and refunds. Periods without sales count as zero. Forecast ranges are the ranges of the chosen model, checked on past data. Associations are not causes.

Data-quality issues

None.

Notes from the analysis

- The first or last week is only partly inside the range, so it looks smaller.

Analysis modules

Module	Status	Reason
Headline figures	Completed	
Churn risk	Completed	
Top contributors	Completed	
Revenue drivers	Completed	

Module	Status	Reason
Forecast	Completed	
Unusual days	Completed	
Customer segments	Completed	
Cohort retention	Completed	
Trend and seasonality	Completed	

Appendix: evidence

The exact queries behind each finding and how many rows each returned.

Finding: Headline figures (KPI table)
Shows: Headline figures for the range and the window before it · **Rows returned:** 2

```
SELECT CASE WHEN "order_date" >= DATE '2024-10-03' THEN 'current' ELSE 'previous' END AS part, count(*) AS rows, sum("revenue") AS net_revenue, sum(CASE WHEN "revenue" > 0 THEN "revenue" END) AS gross_revenue, count(DISTINCT CASE WHEN "revenue" > 0 THEN "order_id" END) AS orders, sum("quantity") AS units, sum(CASE WHEN "cost" IS NOT NULL THEN "revenue" END) AS costed_revenue, sum(CASE WHEN "revenue" IS NOT NULL THEN "cost" END) AS cost FROM data WHERE "revenue" IS NOT NULL AND "order_date" BETWEEN DATE '2024-07-05' AND DATE '2024-12-31' GROUP BY 1 ORDER BY 1
```

Finding: Headline figures (KPI table)
Shows: Net revenue and orders by week · **Rows returned:** 14

```
SELECT CAST(date_trunc('week', "order_date") AS DATE) AS period, count(*) AS rows, sum("revenue") AS net_revenue, sum(CASE WHEN "revenue" > 0 THEN "revenue" END) AS gross_revenue, count(DISTINCT CASE WHEN "revenue" > 0 THEN "order_id" END) AS orders, sum("quantity") AS units, sum(CASE WHEN "cost" IS NOT NULL THEN "revenue" END) AS costed_revenue, sum(CASE WHEN "revenue" IS NOT NULL THEN "cost" END) AS cost FROM data WHERE "revenue" IS NOT NULL AND "order_date" BETWEEN DATE '2024-10-03' AND DATE '2024-12-31' GROUP BY 1 ORDER BY 1
```

Finding: Year to date is 196.7% ahead of last year
Shows: Year-to-date and prior-year-to-date net revenue · **Rows returned:** 1

```
SELECT sum(CASE WHEN "order_date" >= DATE '2024-01-01' AND "order_date" <= DATE '2024-12-31' THEN "revenue" END) AS ytd, sum(CASE WHEN "order_date" >= DATE '2023-01-01' AND "order_date" <= DATE '2023-12-31' THEN "revenue" END) AS prior_ytd, count(*) AS rows FROM data WHERE "revenue" IS NOT NULL AND "order_date" IS NOT NULL
```

Finding: Lasting downward shift in orders from 2024-03-11
Shows: Weekly revenue and orders · **Rows returned:** 103

```
SELECT CAST(date_trunc('week', "order_date") AS DATE) AS week, sum("revenue") AS revenue, count(DISTINCT "order_id") FILTER (WHERE "revenue" > 0) AS orders FROM data WHERE "order_date" IS NOT NULL AND "revenue" IS NOT NULL AND "order_date" >= DATE '2023-01-09' AND "order_date" < DATE '2024-12-30' GROUP BY 1 ORDER BY 1
```

Finding: Net revenue for Furniture is estimated at about 1,339,349.34 next month (80% range: 588,591.74 to 2,090,106.93)
Shows: Net revenue per month for Furniture · **Rows returned:** 24

```
SELECT CAST(date_trunc('month', "order_date") AS DATE) AS period, "category" AS grp, sum("revenue") AS value, count(*) AS rows FROM data WHERE "revenue" IS NOT NULL AND "order_date" IS NOT NULL AND "category" IS NOT NULL AND "category" IN ('Furniture') GROUP BY 1, 2 ORDER BY 2, 1
```

Finding: Net revenue is estimated at about 2,295,213.14 next month (80% range: 1,105,837.97 to 3,484,588.30)
Shows: Net revenue per month · **Rows returned:** 24

```
SELECT CAST(date_trunc('month', "order_date") AS DATE) AS period, sum("revenue") AS value, count(*) AS rows FROM data WHERE "revenue" IS NOT NULL AND "order_date" IS NOT NULL GROUP BY 1 ORDER BY 1
```

Finding: Champions customers bring 50.8% of revenue
Shows: Customers, revenue and averages per RFM segment · **Rows returned:** 7

```
WITH purchases AS (SELECT "customer_id" AS customer, "order_date" AS day, "order_id" AS order_key, "revenue" AS revenue FROM data WHERE "customer_id" IS NOT NULL AND "order_date" IS NOT NULL AND "revenue" > 0), customers AS (SELECT customer, date_diff('day', max(day), DATE '2024-12-31') AS recency_days, count(DISTINCT order_key) AS frequency, sum(revenue) AS monetary, date_diff('day', min(day), DATE '2024-12-31') AS tenure_days FROM purchases GROUP BY customer), scored AS (SELECT *, CAST(1 + floor(5 * (rank() OVER (ORDER BY recency_days DESC) - 1) / count(*) OVER ()) AS INTEGER) AS r_score, CAST(1 + floor(5 * (rank() OVER (ORDER BY frequency ASC) - 1) / count(*) OVER ()) AS INTEGER) AS f_score, CAST(1 + floor(5 * (rank() OVER (ORDER BY monetary ASC) - 1) / count(*) OVER ()) AS INTEGER) AS m_score FROM customers), segmented AS (SELECT *, CASE WHEN r_score = 5 AND f_score = 5 THEN 'Champions' WHEN r_score = 5 AND f_score = 4 THEN 'Champions' WHEN r_score = 5 AND f_score = 3 THEN 'Potential loyalists' WHEN r_score = 5 AND f_score = 2 THEN 'Potential loyalists' WHEN r_score = 5 AND f_score = 1 THEN 'New' WHEN r_score = 4 AND f_score = 5 THEN 'Champions' WHEN r_score = 4 AND f_score = 4 THEN 'Champions' WHEN r_score = 4 AND f_score = 3 THEN 'Potential loyalists' WHEN r_score = 4 AND f_score = 2 THEN 'Potential loyalists' WHEN r_score = 4 AND f_score = 1 THEN 'New' WHEN r_score = 3 AND f_score = 5 THEN 'Loyal' WHEN r_score = 3 AND f_score = 4 THEN 'Loyal' WHEN r_score = 3 AND f_score = 3 THEN 'Loyal' WHEN r_score = 3 AND f_score = 2 THEN 'Hibernating' WHEN r_score = 3 AND f_score = 1 THEN 'Hibernating' WHEN r_score = 2 AND f_score = 5 THEN 'At risk' WHEN r_score = 2 AND f_score = 4 THEN 'At risk' WHEN r_score = 2 AND f_score = 3 THEN 'At risk' WHEN r_score = 2 AND f_score = 2 THEN 'Hibernating' WHEN r_score = 2 AND f_score = 1 THEN 'Hibernating' WHEN r_score = 1 AND f_score = 5 THEN 'At risk' WHEN r_score = 1 AND f_score = 4 THEN 'At risk' WHEN r_score = 1 AND f_score = 3 THEN 'At risk' WHEN r_score = 1 AND f_score = 2 THEN 'Lost' WHEN r_score = 1 AND f_score = 1 THEN 'Lost' END AS segment FROM scored) SELECT segment, count(*) AS customers, round(100.0 * sum(monetary) / sum(sum(monetary)) OVER (), 2) AS revenue_share_pct, round(sum(monetary), 2) AS revenue, round(avg(recency_days), 1) AS avg_recency_days, round(avg(frequency), 2) AS avg_orders, round(avg(monetary), 2) AS avg_spend, round(sum(monetary * 365.0 / greatest(tenure_days, 365)), 2) AS yearly_spend FROM segmented GROUP BY segment ORDER BY revenue DESC, segment
```

Finding: Revenue is up 166.5% vs the same month last year (2024-12)
Shows: Net revenue by month · **Rows returned:** 24

```
SELECT CAST(date_trunc('month', "order_date") AS DATE) AS period, sum("revenue") AS revenue, count(*) AS rows FROM data WHERE "revenue" IS NOT NULL AND "order_date" IS NOT NULL GROUP BY 1 ORDER BY 1
```

Finding: 355 customers are at high estimated churn risk
Shows: Active customers at the training cutoff and how many did not buy again within 32 days · **Rows returned:** 1

```
WITH purchases AS (SELECT "customer_id" AS customer, "order_date" AS day, "order_id" AS order_key, CAST("revenue" AS DOUBLE) AS revenue, "category" AS category, CAST("discount_pct" AS DOUBLE) AS discount FROM data WHERE "customer_id" IS NOT NULL AND "order_date" IS NOT NULL AND "revenue" > 0 AND "order_date" <= DATE '2024-10-28'), active AS (SELECT customer FROM purchases GROUP BY customer HAVING max(day) > DATE '2023-10-29'), future AS (SELECT DISTINCT "customer_id" AS customer FROM data WHERE "customer_id" IS NOT NULL AND "revenue" > 0 AND "order_date" > DATE '2024-10-28' AND "order_date" <= DATE '2024-11-29') SELECT DATE '2024-10-28' AS cutoff, 32 AS window_days, count(*) AS active_customers, count(*) FILTER (WHERE f.customer IS NULL) AS churned, round(100.0 * count(*) FILTER (WHERE f.customer IS NULL) / count(*), 2) AS churn_rate_pct FROM active a LEFT JOIN future f ON a.customer = f.customer
```

Finding: 355 customers are at high estimated churn risk
Shows: Active customers at the evaluation cutoff and how many did not buy again within 32 days · **Rows returned:** 1

```
WITH purchases AS (SELECT "customer_id" AS customer, "order_date" AS day, "order_id" AS order_key, CAST("revenue" AS DOUBLE) AS revenue, "category" AS category, CAST("discount_pct" AS DOUBLE) AS discount FROM data WHERE "customer_id" IS NOT NULL AND "order_date" IS NOT NULL AND "revenue" > 0 AND "order_date" <= DATE '2024-11-29'), active AS (SELECT customer FROM purchases GROUP BY customer HAVING max(day) > DATE '2023-11-30'), future AS (SELECT DISTINCT "customer_id" AS customer FROM data WHERE "customer_id" IS NOT NULL AND "revenue" > 0 AND "order_date" > DATE '2024-11-29' AND "order_date" <= DATE '2024-12-31') SELECT DATE '2024-11-29' AS cutoff, 32 AS window_days, count(*) AS active_customers, count(*) FILTER (WHERE f.customer IS NULL) AS churned, round(100.0 * count(*) FILTER (WHERE f.customer IS NULL) / count(*), 2) AS churn_rate_pct FROM active a LEFT JOIN future f ON a.customer = f.customer
```

Finding: 355 customers are at high estimated churn risk
Shows: Averages of churners (1) and stayers (0) at the evaluation cutoff 2024-11-29 · **Rows returned:** 2

```
WITH base AS (WITH base AS (WITH purchases AS (SELECT "customer_id" AS customer, "order_date" AS day, "order_id" AS order_key, CAST("revenue" AS DOUBLE) AS revenue, "category" AS category, CAST("discount_pct" AS DOUBLE) AS discount FROM data WHERE "customer_id" IS NOT NULL AND "order_date" IS NOT NULL AND "revenue" > 0 AND "order_date" <= DATE '2024-11-29'), feats AS (SELECT customer, date_diff('day', max(day), DATE '2024-11-29') AS recency_days, count(DISTINCT order_key) AS frequency, sum(revenue) AS monetary, date_diff('day', min(day), DATE '2024-11-29') AS tenure_days, coalesce(sum(revenue) FILTER (WHERE day > DATE '2024-08-31'), 0) AS spend_recent, coalesce(sum(revenue) FILTER (WHERE day > DATE '2024-06-02' AND day <= DATE '2024-08-31'), 0) AS spend_before, CAST(date_diff('day', min(day), max(day)) AS DOUBLE) / nullif(count(DISTINCT day) - 1, 0) AS avg_gap_days, count(DISTINCT category) AS category_count, coalesce(sum(revenue) FILTER (WHERE discount > 0), 0) / sum(revenue) AS discount_share FROM purchases GROUP BY customer) SELECT * FROM feats WHERE recency_days < 365 ORDER BY customer), future AS (SELECT DISTINCT "customer_id" AS customer FROM data WHERE "customer_id" IS NOT NULL AND "revenue" > 0 AND "order_date" > DATE '2024-11-29' AND "order_date" <= DATE '2024-12-31') SELECT base.*, CASE WHEN f.customer IS NULL THEN 1 ELSE 0 END AS churned FROM base LEFT JOIN future f ON base.customer = f.customer ORDER BY base.customer) SELECT churned, count(*) AS customers, round(avg(recency_days), 2) AS avg_recency_days, round(avg(frequency), 2) AS avg_orders, round(avg(monetary), 2) AS avg_spend, round(avg(tenure_days), 2) AS avg_tenure_days FROM base GROUP BY churned ORDER BY churned
```

Finding: 4 customer groups found
Shows: Clustering inputs per customer (averages and medians) · **Rows returned:** 1

```
WITH purchases AS (SELECT "customer_id" AS customer, "order_date" AS day, "order_id" AS order_key, "revenue" AS revenue FROM data WHERE "customer_id" IS NOT NULL AND "order_date" IS NOT NULL AND "revenue" > 0), customers AS (SELECT customer, date_diff('day', max(day), DATE '2024-12-31') AS recency_days, count(DISTINCT order_key) AS frequency, sum(revenue) AS monetary, date_diff('day', min(day), DATE '2024-12-31') AS tenure_days FROM purchases GROUP BY customer), scored AS (SELECT *, CAST(1 + floor(5 * (rank() OVER (ORDER BY recency_days DESC) - 1) / count(*) OVER ()) AS INTEGER) AS r_score, CAST(1 + floor(5 * (rank() OVER (ORDER BY frequency ASC) - 1) / count(*) OVER ()) AS INTEGER) AS f_score, CAST(1 + floor(5 * (rank() OVER (ORDER BY monetary ASC) - 1) / count(*) OVER ()) AS INTEGER) AS m_score FROM customers), segmented AS (SELECT *, CASE WHEN r_score = 5 AND f_score = 5 THEN 'Champions' WHEN r_score = 5 AND f_score = 4 THEN 'Champions' WHEN r_score = 5 AND f_score = 3 THEN 'Potential loyalists' WHEN r_score = 5 AND f_score = 2 THEN 'Potential loyalists' WHEN r_score = 5 AND f_score = 1 THEN 'New' WHEN r_score = 4 AND f_score = 5 THEN 'Champions' WHEN r_score = 4 AND f_score = 4 THEN 'Champions' WHEN r_score = 4 AND f_score = 3 THEN 'Potential loyalists' WHEN r_score = 4 AND f_score = 2 THEN 'Potential loyalists' WHEN r_score = 4 AND f_score = 1 THEN 'New' WHEN r_score = 3 AND f_score = 5 THEN 'Loyal' WHEN r_score = 3 AND f_score = 4 THEN 'Loyal' WHEN r_score = 3 AND f_score = 3 THEN 'Loyal' WHEN r_score = 3 AND f_score = 2 THEN 'Hibernating' WHEN r_score = 3 AND f_score = 1 THEN 'Hibernating' WHEN r_score = 2 AND f_score = 5 THEN 'At risk' WHEN r_score = 2 AND f_score = 4 THEN 'At risk' WHEN r_score = 2 AND f_score = 3 THEN 'At risk' WHEN r_score = 2 AND f_score = 2 THEN 'Hibernating' WHEN r_score = 2 AND f_score = 1 THEN 'Hibernating' WHEN r_score = 1 AND f_score = 5 THEN 'At risk' WHEN r_score = 1 AND f_score = 4 THEN 'At risk' WHEN r_score = 1 AND f_score = 3 THEN 'At risk' WHEN r_score = 1 AND f_score = 2 THEN 'Lost' WHEN r_score = 1 AND f_score = 1 THEN 'Lost' END AS segment FROM scored), features AS (SELECT *, monetary / frequency AS aov FROM segmented) SELECT count(*) AS customers, round(avg(recency_days), 2) AS avg_recency_days, round(quantile_cont(recency_days, 0.5), 2) AS median_recency_days, round(avg(frequency), 2) AS avg_frequency, round(quantile_cont(frequency, 0.5), 2) AS median_frequency, round(avg(monetary), 2) AS avg_monetary, round(quantile_cont(monetary, 0.5), 2) AS median_monetary, round(avg(aov), 2) AS avg_aov, round(quantile_cont(aov, 0.5), 2) AS median_aov, round(avg(tenure_days), 2) AS avg_tenure_days, round(quantile_cont(tenure_days, 0.5), 2) AS median_tenure_days FROM features
```

Finding: Net revenue: 12,183,302.00
Shows: Revenue totals: net, gross sales and returns/refunds · **Rows returned:** 1

```
SELECT count(*) AS rows, sum("revenue") AS net_revenue, sum(CASE WHEN "revenue" > 0 THEN "revenue" END) AS gross_revenue, sum(CASE WHEN "revenue" < 0 THEN -"revenue" END) AS returns_total, count(CASE WHEN "revenue" < 0 THEN 1 END) AS returns_count, count(CASE WHEN "revenue" > 0 THEN 1 END) AS sale_rows FROM data WHERE "revenue" IS NOT NULL
```

Finding: Net revenue: 12,183,302.00
Shows: Distinct orders with positive revenue · **Rows returned:** 1

```
SELECT count(DISTINCT "order_id") AS orders FROM data WHERE "revenue" > 0 AND "order_id" IS NOT NULL
```

Finding: Net revenue: 12,183,302.00
Shows: Units sold and returned · **Rows returned:** 1

```
SELECT sum("quantity") AS net_units, sum(CASE WHEN "quantity" > 0 THEN "quantity" END) AS gross_units, sum(CASE WHEN "quantity" < 0 THEN -"quantity" END) AS returned_units FROM data WHERE "quantity" IS NOT NULL
```

Finding: Net revenue: 12,183,302.00
Shows: Revenue and cost on rows that have both · **Rows returned:** 1

```
SELECT sum("revenue") AS revenue, sum("cost") AS cost FROM data WHERE "revenue" IS NOT NULL AND "cost" IS NOT NULL
```

Finding: Peak season: December
Shows: Net revenue by calendar month · **Rows returned:** 24

```
SELECT CAST(date_trunc('month', "order_date") AS DATE) AS period, sum("revenue") AS revenue, count(*) AS rows FROM data WHERE "revenue" IS NOT NULL AND "order_date" IS NOT NULL GROUP BY 1 ORDER BY 1
```

Finding: Net revenue for Electronics is estimated at about 389,080.62 next month (80% range: 157,001.64 to 621,159.59)
Shows: Net revenue per month for Electronics · **Rows returned:** 24

```
SELECT CAST(date_trunc('month', "order_date") AS DATE) AS period, "category" AS grp, sum("revenue") AS value, count(*) AS rows FROM data WHERE "revenue" IS NOT NULL AND "order_date" IS NOT NULL AND "category" IS NOT NULL AND "category" IN ('Electronics') GROUP BY 1, 2 ORDER BY 2, 1
```

Finding: 56.8% of customers return the next month
Shows: Customers per cohort and month offset · **Rows returned:** 300

```
WITH purchases AS (SELECT "customer_id" AS customer, CAST(date_trunc('month', "order_date") AS DATE) AS month FROM data WHERE "customer_id" IS NOT NULL AND "order_date" IS NOT NULL AND "order_date" < DATE '2025-01-01' AND "revenue" > 0 GROUP BY 1, 2), firsts AS (SELECT customer, min(month) AS cohort FROM purchases GROUP BY customer) SELECT f.cohort AS cohort, date_diff('month', f.cohort, p.month) AS month_offset, count(DISTINCT p.customer) AS customers FROM purchases AS p JOIN firsts AS f ON p.customer = f.customer GROUP BY 1, 2 ORDER BY 1, 2
```

Finding: Unit price is the factor most associated with revenue per line
Shows: Average revenue per line by unit price (raw association, no controls) · **Rows returned:** 9

```
SELECT round(CAST("unit_price" AS DOUBLE), 4) AS level, count(*) AS lines, round(avg("revenue"), 2) AS avg_revenue FROM data WHERE "revenue" > 0 AND "order_date" IS NOT NULL AND CAST("unit_price" AS DOUBLE) IS NOT NULL GROUP BY 1 ORDER BY 1 LIMIT 20
```

Finding: Unit price is the factor most associated with revenue per line
Shows: Average revenue per line by discount (raw association, no controls) · **Rows returned:** 6

```
SELECT round(CAST(("discount_pct" / 100.0) AS DOUBLE), 4) AS level, count(*) AS lines, round(avg("revenue"), 2) AS avg_revenue FROM data WHERE "revenue" > 0 AND "order_date" IS NOT NULL AND CAST(("discount_pct" / 100.0) AS DOUBLE) IS NOT NULL GROUP BY 1 ORDER BY 1 LIMIT 20
```

Finding: Unit price is the factor most associated with revenue per line
Shows: Average revenue per line by day of the week (raw association, no controls) · **Rows returned:** 7

```
SELECT dayname("order_date") AS level, count(*) AS lines, round(avg("revenue"), 2) AS avg_revenue FROM data WHERE "revenue" > 0 AND "order_date" IS NOT NULL AND dayname("order_date") IS NOT NULL GROUP BY 1 ORDER BY 1 LIMIT 20
```

Finding: Top category: Furniture (57.2% of sales)
Shows: Positive sales by category · **Rows returned:** 5

```
SELECT "category" AS item, sum("revenue") AS revenue, count(*) AS rows FROM data WHERE "revenue" > 0 AND "category" IS NOT NULL GROUP BY "category" ORDER BY revenue DESC, item
```

Finding: Lasting downward shift in orders from 2024-08-19
Shows: Weekly revenue and orders · **Rows returned:** 103

```
SELECT CAST(date_trunc('week', "order_date") AS DATE) AS week, sum("revenue") AS revenue, count(DISTINCT "order_id") FILTER (WHERE "revenue" > 0) AS orders FROM data WHERE "order_date" IS NOT NULL AND "revenue" IS NOT NULL AND "order_date" >= DATE '2023-01-09' AND "order_date" < DATE '2024-12-30' GROUP BY 1 ORDER BY 1
```

Finding: Net revenue for Grocery is estimated at about 353,512.74 next month (80% range: 161,571.36 to 545,454.13)
Shows: Net revenue per month for Grocery · **Rows returned:** 24

```
SELECT CAST(date_trunc('month', "order_date") AS DATE) AS period, "category" AS grp, sum("revenue") AS value, count(*) AS rows FROM data WHERE "revenue" IS NOT NULL AND "order_date" IS NOT NULL AND "category" IS NOT NULL AND "category" IN ('Grocery') GROUP BY 1, 2 ORDER BY 2, 1
```

Finding: Strongest growth and steepest decline
Shows: Net revenue by month · **Rows returned:** 24

```
SELECT CAST(date_trunc('month', "order_date") AS DATE) AS period, sum("revenue") AS revenue, count(*) AS rows FROM data WHERE "revenue" IS NOT NULL AND "order_date" IS NOT NULL GROUP BY 1 ORDER BY 1
```

Finding: Top region: North (39.4% of sales)
Shows: Positive sales by region · **Rows returned:** 4

```
SELECT "region" AS item, sum("revenue") AS revenue, count(*) AS rows FROM data WHERE "revenue" > 0 AND "region" IS NOT NULL GROUP BY "region" ORDER BY revenue DESC, item
```

Finding: Top channel: Online (33.6% of sales)
Shows: Positive sales by channel · **Rows returned:** 3

```
SELECT "channel" AS item, sum("revenue") AS revenue, count(*) AS rows FROM data WHERE "revenue" > 0 AND "channel" IS NOT NULL GROUP BY "channel" ORDER BY revenue DESC, item
```

Finding: Top product: Item-07 (28.5% of sales)
Shows: Positive sales by product · **Rows returned:** 30

```
SELECT "product" AS item, sum("revenue") AS revenue, count(*) AS rows FROM data WHERE "revenue" > 0 AND "product" IS NOT NULL GROUP BY "product" ORDER BY revenue DESC, item
```

Finding: 8 unusual days found
Shows: Revenue, orders and more per day · **Rows returned:** 713

```
SELECT "order_date" AS period, sum("revenue") AS revenue, count(DISTINCT "order_id") FILTER (WHERE "revenue" > 0) AS orders, coalesce(sum("quantity") FILTER (WHERE "quantity" > 0), 0) AS units, coalesce(avg(("discount_pct" / 100.0)), 0) AS discount_rate FROM data WHERE "order_date" IS NOT NULL AND "revenue" IS NOT NULL GROUP BY 1 ORDER BY 1
```

Finding: Revenue spike on 2024-02-20: 13.3x the expected level
Shows: Revenue per day around the anomaly · **Rows returned:** 29

```
SELECT "order_date" AS period, sum("revenue") AS revenue, count(DISTINCT "order_id") FILTER (WHERE "revenue" > 0) AS orders, coalesce(sum("quantity") FILTER (WHERE "quantity" > 0), 0) AS units, coalesce(avg(("discount_pct" / 100.0)), 0) AS discount_rate FROM data WHERE "order_date" IS NOT NULL AND "revenue" IS NOT NULL AND "order_date" >= DATE '2024-02-06' AND "order_date" < DATE '2024-03-06' GROUP BY 1 ORDER BY 1
```

Finding: Revenue spike on 2024-02-20: 13.3x the expected level
Shows: Likely contributors by product: during vs before · **Rows returned:** 30

```
WITH event AS (SELECT "product" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2024-02-20' AND "order_date" < DATE '2024-02-21' AND "product" IS NOT NULL GROUP BY 1), base AS (SELECT "product" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2024-01-23' AND "order_date" < DATE '2024-02-20' AND "product" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 28, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 28, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day DESC, item
```

Finding: Revenue spike on 2024-02-20: 13.3x the expected level
Shows: Likely contributors by category: during vs before · **Rows returned:** 5

```
WITH event AS (SELECT "category" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2024-02-20' AND "order_date" < DATE '2024-02-21' AND "category" IS NOT NULL GROUP BY 1), base AS (SELECT "category" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2024-01-23' AND "order_date" < DATE '2024-02-20' AND "category" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 28, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 28, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day DESC, item
```

Finding: Revenue spike on 2024-02-20: 13.3x the expected level
Shows: Likely contributors by region: during vs before · **Rows returned:** 4

```
WITH event AS (SELECT "region" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2024-02-20' AND "order_date" < DATE '2024-02-21' AND "region" IS NOT NULL GROUP BY 1), base AS (SELECT "region" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2024-01-23' AND "order_date" < DATE '2024-02-20' AND "region" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 28, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 28, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day DESC, item
```

Finding: Revenue spike on 2024-02-20: 13.3x the expected level
Shows: Likely contributors by channel: during vs before · **Rows returned:** 3

```
WITH event AS (SELECT "channel" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2024-02-20' AND "order_date" < DATE '2024-02-21' AND "channel" IS NOT NULL GROUP BY 1), base AS (SELECT "channel" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2024-01-23' AND "order_date" < DATE '2024-02-20' AND "channel" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 28, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 28, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day DESC, item
```

Finding: Strongest day: Tuesday
Shows: Net revenue and trading days by weekday · **Rows returned:** 7

```
SELECT dayofweek("order_date") AS weekday, sum("revenue") AS revenue, count(DISTINCT "order_date") AS days, count(*) AS rows FROM data WHERE "revenue" IS NOT NULL AND "order_date" IS NOT NULL GROUP BY 1 ORDER BY 1
```

Finding: 2 products with no sales in the last 90 days
Shows: Products with no recent sales · **Rows returned:** 2

```
WITH quiet AS (SELECT "product" AS item, max("order_date") AS last_sale FROM data WHERE "revenue" > 0 AND "order_date" IS NOT NULL AND "product" IS NOT NULL GROUP BY "product" HAVING max("order_date") < DATE '2024-10-02') SELECT q.item AS item, q.last_sale AS last_sale, sum(x."revenue") AS lifetime_revenue, sum(CASE WHEN x."order_date" > q.last_sale - 90 THEN x."revenue" END) AS run_rate FROM data AS x JOIN quiet AS q ON x."product" = q.item WHERE x."revenue" > 0 AND x."order_date" IS NOT NULL GROUP BY q.item, q.last_sale ORDER BY lifetime_revenue DESC, item
```

Finding: Revenue spike on 2023-03-14: 21.2x the expected level
Shows: Revenue per day around the anomaly · **Rows returned:** 29

```
SELECT "order_date" AS period, sum("revenue") AS revenue, count(DISTINCT "order_id") FILTER (WHERE "revenue" > 0) AS orders, coalesce(sum("quantity") FILTER (WHERE "quantity" > 0), 0) AS units, coalesce(avg(("discount_pct" / 100.0)), 0) AS discount_rate FROM data WHERE "order_date" IS NOT NULL AND "revenue" IS NOT NULL AND "order_date" >= DATE '2023-02-28' AND "order_date" < DATE '2023-03-29' GROUP BY 1 ORDER BY 1
```

Finding: Revenue spike on 2023-03-14: 21.2x the expected level
Shows: Likely contributors by product: during vs before · **Rows returned:** 30

```
WITH event AS (SELECT "product" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-03-14' AND "order_date" < DATE '2023-03-15' AND "product" IS NOT NULL GROUP BY 1), base AS (SELECT "product" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-02-14' AND "order_date" < DATE '2023-03-14' AND "order_date" NOT IN (DATE '2023-02-17') AND "product" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 27, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 27, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day DESC, item
```

Finding: Revenue spike on 2023-03-14: 21.2x the expected level
Shows: Likely contributors by category: during vs before · **Rows returned:** 5

```
WITH event AS (SELECT "category" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-03-14' AND "order_date" < DATE '2023-03-15' AND "category" IS NOT NULL GROUP BY 1), base AS (SELECT "category" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-02-14' AND "order_date" < DATE '2023-03-14' AND "order_date" NOT IN (DATE '2023-02-17') AND "category" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 27, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 27, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day DESC, item
```

Finding: Revenue spike on 2023-03-14: 21.2x the expected level
Shows: Likely contributors by region: during vs before · **Rows returned:** 4

```
WITH event AS (SELECT "region" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-03-14' AND "order_date" < DATE '2023-03-15' AND "region" IS NOT NULL GROUP BY 1), base AS (SELECT "region" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-02-14' AND "order_date" < DATE '2023-03-14' AND "order_date" NOT IN (DATE '2023-02-17') AND "region" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 27, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 27, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day DESC, item
```

Finding: Revenue spike on 2023-03-14: 21.2x the expected level
Shows: Likely contributors by channel: during vs before · **Rows returned:** 3

```
WITH event AS (SELECT "channel" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-03-14' AND "order_date" < DATE '2023-03-15' AND "channel" IS NOT NULL GROUP BY 1), base AS (SELECT "channel" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-02-14' AND "order_date" < DATE '2023-03-14' AND "order_date" NOT IN (DATE '2023-02-17') AND "channel" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 27, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 27, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day DESC, item
```

Finding: Net revenue for Apparel is estimated at about 26,532.15 next month (80% range: 18,709.89 to 36,028.48)
Shows: Net revenue per month for Apparel · **Rows returned:** 24

```
SELECT CAST(date_trunc('month', "order_date") AS DATE) AS period, "category" AS grp, sum("revenue") AS value, count(*) AS rows FROM data WHERE "revenue" IS NOT NULL AND "order_date" IS NOT NULL AND "category" IS NOT NULL AND "category" IN ('Apparel') GROUP BY 1, 2 ORDER BY 2, 1
```

Finding: Net revenue for Office is estimated at about 34,741.62 next month (80% range: 15,672.73 to 53,810.52)
Shows: Net revenue per month for Office · **Rows returned:** 24

```
SELECT CAST(date_trunc('month', "order_date") AS DATE) AS period, "category" AS grp, sum("revenue") AS value, count(*) AS rows FROM data WHERE "revenue" IS NOT NULL AND "order_date" IS NOT NULL AND "category" IS NOT NULL AND "category" IN ('Office') GROUP BY 1, 2 ORDER BY 2, 1
```

Finding: Revenue drop on 2023-05-09: 99.5% below the expected level
Shows: Revenue per day around the anomaly · **Rows returned:** 29

```
SELECT "order_date" AS period, sum("revenue") AS revenue, count(DISTINCT "order_id") FILTER (WHERE "revenue" > 0) AS orders, coalesce(sum("quantity") FILTER (WHERE "quantity" > 0), 0) AS units, coalesce(avg(("discount_pct" / 100.0)), 0) AS discount_rate FROM data WHERE "order_date" IS NOT NULL AND "revenue" IS NOT NULL AND "order_date" >= DATE '2023-04-25' AND "order_date" < DATE '2023-05-24' GROUP BY 1 ORDER BY 1
```

Finding: Revenue drop on 2023-05-09: 99.5% below the expected level
Shows: Likely contributors by product: during vs before · **Rows returned:** 28

```
WITH event AS (SELECT "product" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-05-09' AND "order_date" < DATE '2023-05-10' AND "product" IS NOT NULL GROUP BY 1), base AS (SELECT "product" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-04-11' AND "order_date" < DATE '2023-05-09' AND "product" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 28, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 28, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day ASC, item
```

Finding: Revenue drop on 2023-05-09: 99.5% below the expected level
Shows: Likely contributors by category: during vs before · **Rows returned:** 5

```
WITH event AS (SELECT "category" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-05-09' AND "order_date" < DATE '2023-05-10' AND "category" IS NOT NULL GROUP BY 1), base AS (SELECT "category" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-04-11' AND "order_date" < DATE '2023-05-09' AND "category" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 28, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 28, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day ASC, item
```

Finding: Revenue drop on 2023-05-09: 99.5% below the expected level
Shows: Likely contributors by region: during vs before · **Rows returned:** 4

```
WITH event AS (SELECT "region" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-05-09' AND "order_date" < DATE '2023-05-10' AND "region" IS NOT NULL GROUP BY 1), base AS (SELECT "region" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-04-11' AND "order_date" < DATE '2023-05-09' AND "region" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 28, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 28, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day ASC, item
```

Finding: Revenue drop on 2023-05-09: 99.5% below the expected level
Shows: Likely contributors by channel: during vs before · **Rows returned:** 3

```
WITH event AS (SELECT "channel" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-05-09' AND "order_date" < DATE '2023-05-10' AND "channel" IS NOT NULL GROUP BY 1), base AS (SELECT "channel" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-04-11' AND "order_date" < DATE '2023-05-09' AND "channel" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 28, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 28, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day ASC, item
```

Finding: Revenue drop on 2023-10-17: 99.8% below the expected level
Shows: Revenue per day around the anomaly · **Rows returned:** 29

```
SELECT "order_date" AS period, sum("revenue") AS revenue, count(DISTINCT "order_id") FILTER (WHERE "revenue" > 0) AS orders, coalesce(sum("quantity") FILTER (WHERE "quantity" > 0), 0) AS units, coalesce(avg(("discount_pct" / 100.0)), 0) AS discount_rate FROM data WHERE "order_date" IS NOT NULL AND "revenue" IS NOT NULL AND "order_date" >= DATE '2023-10-03' AND "order_date" < DATE '2023-11-01' GROUP BY 1 ORDER BY 1
```

Finding: Revenue drop on 2023-10-17: 99.8% below the expected level
Shows: Likely contributors by product: during vs before · **Rows returned:** 30

```
WITH event AS (SELECT "product" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-10-17' AND "order_date" < DATE '2023-10-18' AND "product" IS NOT NULL GROUP BY 1), base AS (SELECT "product" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-09-19' AND "order_date" < DATE '2023-10-17' AND "product" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 28, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 28, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day ASC, item
```

Finding: Revenue drop on 2023-10-17: 99.8% below the expected level
Shows: Likely contributors by category: during vs before · **Rows returned:** 5

```
WITH event AS (SELECT "category" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-10-17' AND "order_date" < DATE '2023-10-18' AND "category" IS NOT NULL GROUP BY 1), base AS (SELECT "category" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-09-19' AND "order_date" < DATE '2023-10-17' AND "category" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 28, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 28, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day ASC, item
```

Finding: Revenue drop on 2023-10-17: 99.8% below the expected level
Shows: Likely contributors by region: during vs before · **Rows returned:** 4

```
WITH event AS (SELECT "region" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-10-17' AND "order_date" < DATE '2023-10-18' AND "region" IS NOT NULL GROUP BY 1), base AS (SELECT "region" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-09-19' AND "order_date" < DATE '2023-10-17' AND "region" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 28, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 28, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day ASC, item
```

Finding: Revenue drop on 2023-10-17: 99.8% below the expected level
Shows: Likely contributors by channel: during vs before · **Rows returned:** 3

```
WITH event AS (SELECT "channel" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-10-17' AND "order_date" < DATE '2023-10-18' AND "channel" IS NOT NULL GROUP BY 1), base AS (SELECT "channel" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-09-19' AND "order_date" < DATE '2023-10-17' AND "channel" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 28, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 28, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day ASC, item
```

Finding: Revenue spike on 2023-01-09: 7.2x the expected level
Shows: Revenue per day around the anomaly · **Rows returned:** 11

```
SELECT "order_date" AS period, sum("revenue") AS revenue, count(DISTINCT "order_id") FILTER (WHERE "revenue" > 0) AS orders, coalesce(sum("quantity") FILTER (WHERE "quantity" > 0), 0) AS units, coalesce(avg(("discount_pct" / 100.0)), 0) AS discount_rate FROM data WHERE "order_date" IS NOT NULL AND "revenue" IS NOT NULL AND "order_date" >= DATE '2023-01-06' AND "order_date" < DATE '2023-01-24' GROUP BY 1 ORDER BY 1
```

Finding: Revenue spike on 2023-01-09: 7.2x the expected level
Shows: Likely contributors by product: during vs before · **Rows returned:** 2

```
WITH event AS (SELECT "product" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-09' AND "order_date" < DATE '2023-01-10' AND "product" IS NOT NULL GROUP BY 1), base AS (SELECT "product" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2022-12-12' AND "order_date" < DATE '2023-01-09' AND "product" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 28, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 28, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day DESC, item
```

Finding: Revenue spike on 2023-01-09: 7.2x the expected level
Shows: Likely contributors by category: during vs before · **Rows returned:** 2

```
WITH event AS (SELECT "category" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-09' AND "order_date" < DATE '2023-01-10' AND "category" IS NOT NULL GROUP BY 1), base AS (SELECT "category" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2022-12-12' AND "order_date" < DATE '2023-01-09' AND "category" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 28, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 28, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day DESC, item
```

Finding: Revenue spike on 2023-01-09: 7.2x the expected level
Shows: Likely contributors by region: during vs before · **Rows returned:** 2

```
WITH event AS (SELECT "region" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-09' AND "order_date" < DATE '2023-01-10' AND "region" IS NOT NULL GROUP BY 1), base AS (SELECT "region" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2022-12-12' AND "order_date" < DATE '2023-01-09' AND "region" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 28, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 28, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day DESC, item
```

Finding: Revenue spike on 2023-01-09: 7.2x the expected level
Shows: Likely contributors by channel: during vs before · **Rows returned:** 1

```
WITH event AS (SELECT "channel" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-09' AND "order_date" < DATE '2023-01-10' AND "channel" IS NOT NULL GROUP BY 1), base AS (SELECT "channel" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2022-12-12' AND "order_date" < DATE '2023-01-09' AND "channel" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 28, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 28, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day DESC, item
```

Finding: Revenue spike on 2023-01-29: 106.3x the expected level
Shows: Revenue per day around the anomaly · **Rows returned:** 22

```
SELECT "order_date" AS period, sum("revenue") AS revenue, count(DISTINCT "order_id") FILTER (WHERE "revenue" > 0) AS orders, coalesce(sum("quantity") FILTER (WHERE "quantity" > 0), 0) AS units, coalesce(avg(("discount_pct" / 100.0)), 0) AS discount_rate FROM data WHERE "order_date" IS NOT NULL AND "revenue" IS NOT NULL AND "order_date" >= DATE '2023-01-15' AND "order_date" < DATE '2023-02-13' GROUP BY 1 ORDER BY 1
```

Finding: Revenue spike on 2023-01-29: 106.3x the expected level
Shows: Likely contributors by product: during vs before · **Rows returned:** 17

```
WITH event AS (SELECT "product" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-29' AND "order_date" < DATE '2023-01-30' AND "product" IS NOT NULL GROUP BY 1), base AS (SELECT "product" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-01' AND "order_date" < DATE '2023-01-29' AND "order_date" NOT IN (DATE '2023-01-09', DATE '2023-01-27') AND "product" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 26, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 26, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day DESC, item
```

Finding: Revenue spike on 2023-01-29: 106.3x the expected level
Shows: Likely contributors by category: during vs before · **Rows returned:** 5

```
WITH event AS (SELECT "category" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-29' AND "order_date" < DATE '2023-01-30' AND "category" IS NOT NULL GROUP BY 1), base AS (SELECT "category" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-01' AND "order_date" < DATE '2023-01-29' AND "order_date" NOT IN (DATE '2023-01-09', DATE '2023-01-27') AND "category" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 26, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 26, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day DESC, item
```

Finding: Revenue spike on 2023-01-29: 106.3x the expected level
Shows: Likely contributors by region: during vs before · **Rows returned:** 4

```
WITH event AS (SELECT "region" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-29' AND "order_date" < DATE '2023-01-30' AND "region" IS NOT NULL GROUP BY 1), base AS (SELECT "region" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-01' AND "order_date" < DATE '2023-01-29' AND "order_date" NOT IN (DATE '2023-01-09', DATE '2023-01-27') AND "region" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 26, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 26, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day DESC, item
```

Finding: Revenue spike on 2023-01-29: 106.3x the expected level
Shows: Likely contributors by channel: during vs before · **Rows returned:** 3

```
WITH event AS (SELECT "channel" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-29' AND "order_date" < DATE '2023-01-30' AND "channel" IS NOT NULL GROUP BY 1), base AS (SELECT "channel" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-01' AND "order_date" < DATE '2023-01-29' AND "order_date" NOT IN (DATE '2023-01-09', DATE '2023-01-27') AND "channel" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 26, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 26, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day DESC, item
```

Finding: 15 unusual rows found
Shows: Row-level checks over every row · **Rows returned:** 1

```
WITH typical AS (SELECT "product" AS typical_item, median("unit_price") AS typical_price FROM data WHERE "unit_price" > 0 GROUP BY 1) SELECT count(*) FILTER (WHERE "quantity" = 0) AS zero_quantity, count(*) FILTER (WHERE "quantity" < 0 AND "revenue" > 0) AS negative_quantity_positive_revenue, count(*) FILTER (WHERE ("discount_pct" / 100.0) > 0.5) AS unusual_discount, count(*) FILTER (WHERE "quantity" < 0 AND "revenue" < 0) AS returns, count(*) FILTER (WHERE data."unit_price" > 0 AND (data."unit_price" >= 3.0 * typical.typical_price OR data."unit_price" <= typical.typical_price / 3.0)) AS extreme_price, count(*) AS rows FROM data LEFT JOIN typical ON data."product" = typical.typical_item
```

Finding: Churn model check: AUC 0.88 on held-out customers
Shows: Active customers at the training cutoff and how many did not buy again within 32 days · **Rows returned:** 1

```
WITH purchases AS (SELECT "customer_id" AS customer, "order_date" AS day, "order_id" AS order_key, CAST("revenue" AS DOUBLE) AS revenue, "category" AS category, CAST("discount_pct" AS DOUBLE) AS discount FROM data WHERE "customer_id" IS NOT NULL AND "order_date" IS NOT NULL AND "revenue" > 0 AND "order_date" <= DATE '2024-10-28'), active AS (SELECT customer FROM purchases GROUP BY customer HAVING max(day) > DATE '2023-10-29'), future AS (SELECT DISTINCT "customer_id" AS customer FROM data WHERE "customer_id" IS NOT NULL AND "revenue" > 0 AND "order_date" > DATE '2024-10-28' AND "order_date" <= DATE '2024-11-29') SELECT DATE '2024-10-28' AS cutoff, 32 AS window_days, count(*) AS active_customers, count(*) FILTER (WHERE f.customer IS NULL) AS churned, round(100.0 * count(*) FILTER (WHERE f.customer IS NULL) / count(*), 2) AS churn_rate_pct FROM active a LEFT JOIN future f ON a.customer = f.customer
```

Finding: Churn model check: AUC 0.88 on held-out customers
Shows: Active customers at the evaluation cutoff and how many did not buy again within 32 days · **Rows returned:** 1

```
WITH purchases AS (SELECT "customer_id" AS customer, "order_date" AS day, "order_id" AS order_key, CAST("revenue" AS DOUBLE) AS revenue, "category" AS category, CAST("discount_pct" AS DOUBLE) AS discount FROM data WHERE "customer_id" IS NOT NULL AND "order_date" IS NOT NULL AND "revenue" > 0 AND "order_date" <= DATE '2024-11-29'), active AS (SELECT customer FROM purchases GROUP BY customer HAVING max(day) > DATE '2023-11-30'), future AS (SELECT DISTINCT "customer_id" AS customer FROM data WHERE "customer_id" IS NOT NULL AND "revenue" > 0 AND "order_date" > DATE '2024-11-29' AND "order_date" <= DATE '2024-12-31') SELECT DATE '2024-11-29' AS cutoff, 32 AS window_days, count(*) AS active_customers, count(*) FILTER (WHERE f.customer IS NULL) AS churned, round(100.0 * count(*) FILTER (WHERE f.customer IS NULL) / count(*), 2) AS churn_rate_pct FROM active a LEFT JOIN future f ON a.customer = f.customer
```

Finding: Churn model check: AUC 0.88 on held-out customers
Shows: Averages of churners (1) and stayers (0) at the evaluation cutoff 2024-11-29 · **Rows returned:** 2

```
WITH base AS (WITH base AS (WITH purchases AS (SELECT "customer_id" AS customer, "order_date" AS day, "order_id" AS order_key, CAST("revenue" AS DOUBLE) AS revenue, "category" AS category, CAST("discount_pct" AS DOUBLE) AS discount FROM data WHERE "customer_id" IS NOT NULL AND "order_date" IS NOT NULL AND "revenue" > 0 AND "order_date" <= DATE '2024-11-29'), feats AS (SELECT customer, date_diff('day', max(day), DATE '2024-11-29') AS recency_days, count(DISTINCT order_key) AS frequency, sum(revenue) AS monetary, date_diff('day', min(day), DATE '2024-11-29') AS tenure_days, coalesce(sum(revenue) FILTER (WHERE day > DATE '2024-08-31'), 0) AS spend_recent, coalesce(sum(revenue) FILTER (WHERE day > DATE '2024-06-02' AND day <= DATE '2024-08-31'), 0) AS spend_before, CAST(date_diff('day', min(day), max(day)) AS DOUBLE) / nullif(count(DISTINCT day) - 1, 0) AS avg_gap_days, count(DISTINCT category) AS category_count, coalesce(sum(revenue) FILTER (WHERE discount > 0), 0) / sum(revenue) AS discount_share FROM purchases GROUP BY customer) SELECT * FROM feats WHERE recency_days < 365 ORDER BY customer), future AS (SELECT DISTINCT "customer_id" AS customer FROM data WHERE "customer_id" IS NOT NULL AND "revenue" > 0 AND "order_date" > DATE '2024-11-29' AND "order_date" <= DATE '2024-12-31') SELECT base.*, CASE WHEN f.customer IS NULL THEN 1 ELSE 0 END AS churned FROM base LEFT JOIN future f ON base.customer = f.customer ORDER BY base.customer) SELECT churned, count(*) AS customers, round(avg(recency_days), 2) AS avg_recency_days, round(avg(frequency), 2) AS avg_orders, round(avg(monetary), 2) AS avg_spend, round(avg(tenure_days), 2) AS avg_tenure_days FROM base GROUP BY churned ORDER BY churned
```

Finding: Revenue drop on 2023-01-27: 100.0% below the expected level
Shows: Revenue per day around the anomaly · **Rows returned:** 22

```
SELECT "order_date" AS period, sum("revenue") AS revenue, count(DISTINCT "order_id") FILTER (WHERE "revenue" > 0) AS orders, coalesce(sum("quantity") FILTER (WHERE "quantity" > 0), 0) AS units, coalesce(avg(("discount_pct" / 100.0)), 0) AS discount_rate FROM data WHERE "order_date" IS NOT NULL AND "revenue" IS NOT NULL AND "order_date" >= DATE '2023-01-13' AND "order_date" < DATE '2023-02-11' GROUP BY 1 ORDER BY 1
```

Finding: Revenue drop on 2023-01-27: 100.0% below the expected level
Shows: Likely contributors by product: during vs before · **Rows returned:** 15

```
WITH event AS (SELECT "product" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-27' AND "order_date" < DATE '2023-01-28' AND "product" IS NOT NULL GROUP BY 1), base AS (SELECT "product" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2022-12-30' AND "order_date" < DATE '2023-01-27' AND "order_date" NOT IN (DATE '2023-01-09') AND "product" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 27, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 27, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day ASC, item
```

Finding: Revenue drop on 2023-01-27: 100.0% below the expected level
Shows: Likely contributors by category: during vs before · **Rows returned:** 5

```
WITH event AS (SELECT "category" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-27' AND "order_date" < DATE '2023-01-28' AND "category" IS NOT NULL GROUP BY 1), base AS (SELECT "category" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2022-12-30' AND "order_date" < DATE '2023-01-27' AND "order_date" NOT IN (DATE '2023-01-09') AND "category" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 27, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 27, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day ASC, item
```

Finding: Revenue drop on 2023-01-27: 100.0% below the expected level
Shows: Likely contributors by region: during vs before · **Rows returned:** 4

```
WITH event AS (SELECT "region" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-27' AND "order_date" < DATE '2023-01-28' AND "region" IS NOT NULL GROUP BY 1), base AS (SELECT "region" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2022-12-30' AND "order_date" < DATE '2023-01-27' AND "order_date" NOT IN (DATE '2023-01-09') AND "region" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 27, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 27, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day ASC, item
```

Finding: Revenue drop on 2023-01-27: 100.0% below the expected level
Shows: Likely contributors by channel: during vs before · **Rows returned:** 3

```
WITH event AS (SELECT "channel" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-27' AND "order_date" < DATE '2023-01-28' AND "channel" IS NOT NULL GROUP BY 1), base AS (SELECT "channel" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2022-12-30' AND "order_date" < DATE '2023-01-27' AND "order_date" NOT IN (DATE '2023-01-09') AND "channel" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 27, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 27, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day ASC, item
```

Finding: Revenue drop on 2023-02-17: 100.0% below the expected level
Shows: Revenue per day around the anomaly · **Rows returned:** 28

```
SELECT "order_date" AS period, sum("revenue") AS revenue, count(DISTINCT "order_id") FILTER (WHERE "revenue" > 0) AS orders, coalesce(sum("quantity") FILTER (WHERE "quantity" > 0), 0) AS units, coalesce(avg(("discount_pct" / 100.0)), 0) AS discount_rate FROM data WHERE "order_date" IS NOT NULL AND "revenue" IS NOT NULL AND "order_date" >= DATE '2023-02-03' AND "order_date" < DATE '2023-03-04' GROUP BY 1 ORDER BY 1
```

Finding: Revenue drop on 2023-02-17: 100.0% below the expected level
Shows: Likely contributors by product: during vs before · **Rows returned:** 23

```
WITH event AS (SELECT "product" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-02-17' AND "order_date" < DATE '2023-02-18' AND "product" IS NOT NULL GROUP BY 1), base AS (SELECT "product" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-20' AND "order_date" < DATE '2023-02-17' AND "order_date" NOT IN (DATE '2023-01-27', DATE '2023-01-29') AND "product" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 26, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 26, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day ASC, item
```

Finding: Revenue drop on 2023-02-17: 100.0% below the expected level
Shows: Likely contributors by category: during vs before · **Rows returned:** 5

```
WITH event AS (SELECT "category" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-02-17' AND "order_date" < DATE '2023-02-18' AND "category" IS NOT NULL GROUP BY 1), base AS (SELECT "category" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-20' AND "order_date" < DATE '2023-02-17' AND "order_date" NOT IN (DATE '2023-01-27', DATE '2023-01-29') AND "category" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 26, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 26, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day ASC, item
```

Finding: Revenue drop on 2023-02-17: 100.0% below the expected level
Shows: Likely contributors by region: during vs before · **Rows returned:** 4

```
WITH event AS (SELECT "region" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-02-17' AND "order_date" < DATE '2023-02-18' AND "region" IS NOT NULL GROUP BY 1), base AS (SELECT "region" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-20' AND "order_date" < DATE '2023-02-17' AND "order_date" NOT IN (DATE '2023-01-27', DATE '2023-01-29') AND "region" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 26, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 26, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day ASC, item
```

Finding: Revenue drop on 2023-02-17: 100.0% below the expected level
Shows: Likely contributors by channel: during vs before · **Rows returned:** 3

```
WITH event AS (SELECT "channel" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-02-17' AND "order_date" < DATE '2023-02-18' AND "channel" IS NOT NULL GROUP BY 1), base AS (SELECT "channel" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-20' AND "order_date" < DATE '2023-02-17' AND "order_date" NOT IN (DATE '2023-01-27', DATE '2023-01-29') AND "channel" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 26, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 26, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day ASC, item
```